# Istanbul Baraj Projesi - Guncel Durum ve Cikti Ozeti (2026-03-18)

## Ozet

Bu rapor, mevcut model gelistirme durumunu, son test sonuclarini, uretmis oldugumuz ana ciktilari ve guncel deneyleri tek belgede toplar. Egitim-test kurgu su sekildedir: 2015 sonuna kadar egitim, 2016-2020 holdout test.

Yeni odak: iklim ciktilarini (sicaklik, yagis, nem) netlestirip ET0 (FAO Penman-Monteith) ve basit su dengesi (giren-cikan su) ile proje omurgasini kurmak. Insan davranisi ve isletme kararlari bu turda modelin disinda tutuluyor.

## En Iyi Modeller (Holdout 2016-2020)

- hybrid_physics_stacked_exog: MAPE 9.52%, RMSE 6.58, Pearson 94.22%
- quantile_regressor_plus: MAPE 9.86%, RMSE 7.78, Pearson 91.40%
- hybrid_physics_ensemble_phys_old: MAPE 10.47%, RMSE 7.77, Pearson 93.24%
- water_balance_v4_corrected: MAPE 11.55%, RMSE 8.70, Pearson 94.43%
- prophet_regressor_exog: MAPE 24.92%, RMSE 17.96, Pearson 61.88%

## Holdout Ozet Tablosu (ilk 10 satir)

| model | rmse_pp | mae_pp | mape_pct | smape_pct | pearson_corr_pct | spearm |
|---|---|---|---|---|---|---|
| hybrid_physics_stacked_exog | 6.580649091139963 | 5.311002662707452 | 9.523445322804566 | 10.043801978804352 | 94.21532642248675 | 95.5043 |
| quantile_regressor_plus | 7.781408402610931 | 5.9008026283590285 | 9.859026669987475 | 9.976279907648932 | 91.4027488985479 | 90.6751 |
| hybrid_physics_ensemble_phys_old | 7.773890522819784 | 6.000657070913325 | 10.474449077252649 | 9.828667957597132 | 93.24176889774876 | 93.4661 |
| water_balance_v4_corrected | 8.703889909998326 | 6.897682747074565 | 11.551826571743634 | 10.730825184658974 | 94.43183531111052 | 94.6033 |
| prophet_regressor_exog | 17.96326300551316 | 14.358897954591082 | 24.921312563486534 | 27.71533796937452 | 61.87548387471165 | 61.0502 |

## 2040 Sonu Baz Senaryo Uc Degerleri

| model | pred_fill | pred_fill_pct |
|---|---|---|
| hybrid_physics_stacked_exog | 0.1492568561494105 | 14.92568561494105 |
| hybrid_physics_ensemble_phys_old | 0.3578001102160605 | 35.78001102160605 |
| water_balance_v4_corrected | 0.029021234582677 | 2.9021234582677056 |
| quantile_regressor_plus | 0.4764826421385766 | 47.648264213857665 |
| prophet_regressor_exog | 0.391468949390547 | 39.1468949390547 |

## MAPE 5 Hedefi Ic■n Kisa Holdout Direkt Arama

Bu arama, kisa vade one-step dogruluk icin agresif blend aramasidir. Uzun vadeli projeksiyon omurgasi degildir.

| model | rmse_pp | mae_pp | mape_pct | smape_pct | pearson_corr_pct | spearman_cor |
|---|---|---|---|---|---|---|
| evo_direct_level_exog_blend | 4.898806673277172 | 3.2729485823161952 | 5.617946651410206 | 5.610397597107901 | 96.43813167915712 | 95.548763545 |
| gbr_d2_lr002 | 5.835730868647406 | 4.459084450438441 | 7.972740485690652 | 7.593334811226747 | 96.43813167915712 | 95.548763545 |
| gbr_d3_lr0015 | 5.937048324227103 | 4.458547130020621 | 8.390930516034897 | 7.849258430117942 | 96.12766056588434 | 94.848569046 |
| et_mid | 5.887215651848968 | 4.720859389190134 | 9.11386601884326 | 8.448476570419615 | 96.57152450606982 | 95.520978049 |
| et_deep | 6.019703275706171 | 4.802070785316378 | 9.2834152990556 | 8.59882428254586 | 96.32190153214977 | 95.226451792 |

| | | | | | | |
|---|---|---|---|---|---|---|
| hgbr_mid | 6.917888051989912 | 5.369229318895122 | 9.591472958121193 | 9.18850586629846 | 93.57293051836812 | 92.570158377 |
| catboost_d4_lr003 | 6.103881647774073 | 4.795158555150639 | 9.909095907308489 | 8.955019336016942 | 97.68090869725783 | 96.732425673 |
| catboost_d4_lr002 | 6.509189053352535 | 5.075015754072004 | 10.571272650410764 | 9.466119510775805 | 97.68502640670744 | 96.460127813 |
| catboost_d5_lr002 | 6.852003459064616 | 5.4120823259857875 | 11.10418084642938 | 9.942092456956717 | 97.35709340062468 | 95.793275909 |
| catboost_d6_lr015 | 7.904935784760711 | 6.261104291300097 | 12.930757983201197 | 11.401229874915678 | 96.42212554795806 | 94.248402333 |

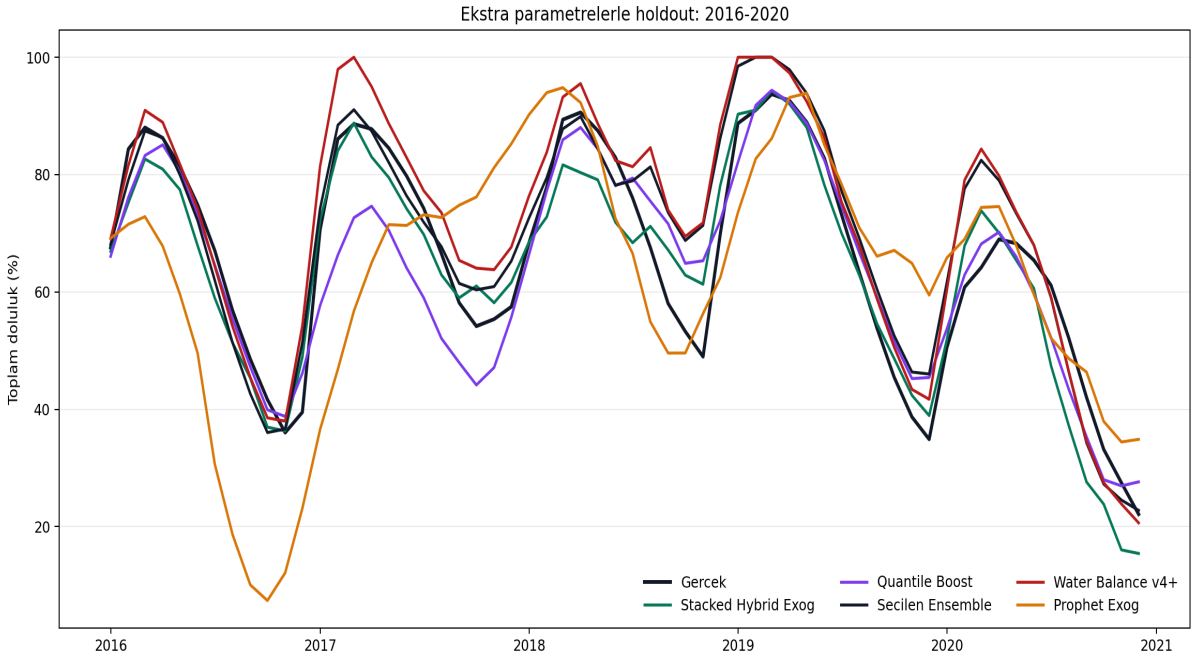
## Water-Balance Denemeleri (Round 5)

| model | rmse_pp | mape_pct | pearson_pct | base_2040_pct | decision |
|---|---|---|---|---|---|
| water_balance_v4_corrected | 8.70389 | 11.551827 | 94.431835 | 2.902123 | baseline_wb_best_so_far |
| water_balance_v6_ops_curve | 11.363206918641266 | 15.409930326322003 | 92.3282992532975 | 6.213523530421872 | reject |
| water_balance_v7_ops_residual | 10.631368325062382 | 13.778137277006728 | 93.16868103216356 | 28.96952858814955 | reject_but_less_bad_than_v6 |

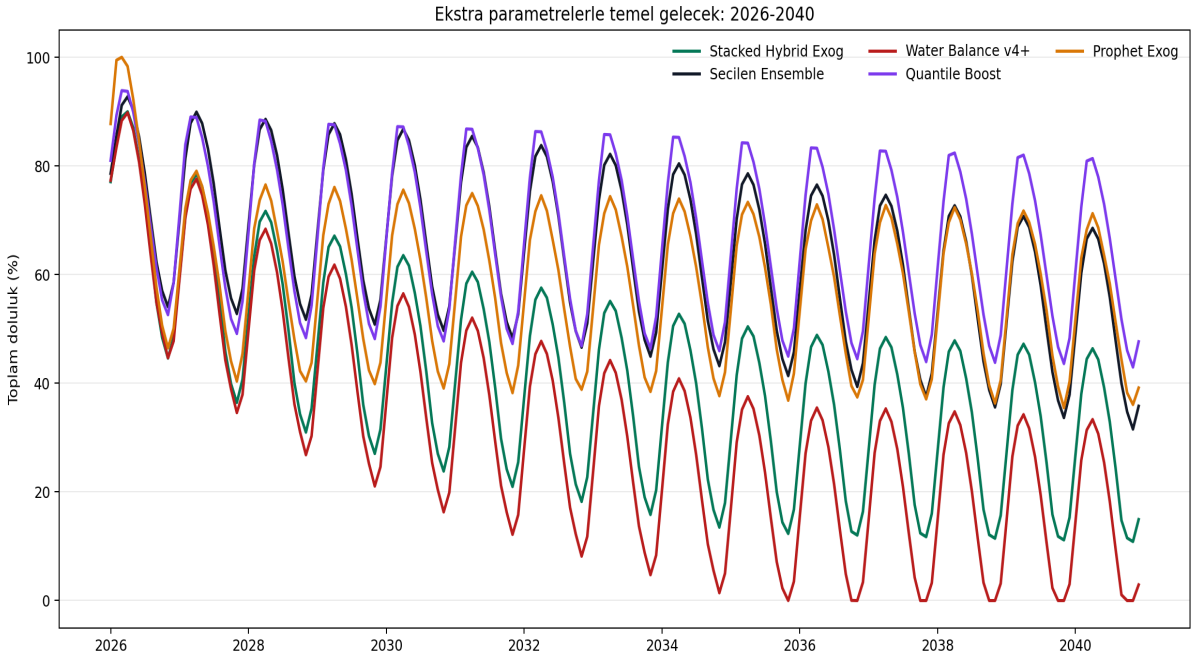
# Ana Cikti Klasorleri

- output/final_delivery/istanbul_baraj_final_paket - output/istanbul_models_extra_params_2026_03_12 - output/istanbul_mape5_push_and_longhorizon_2026_03_12 - output/istanbul_hybrid_physics_sourceaware_ensemble_2040 - output/istanbul_preferred_projection_2040

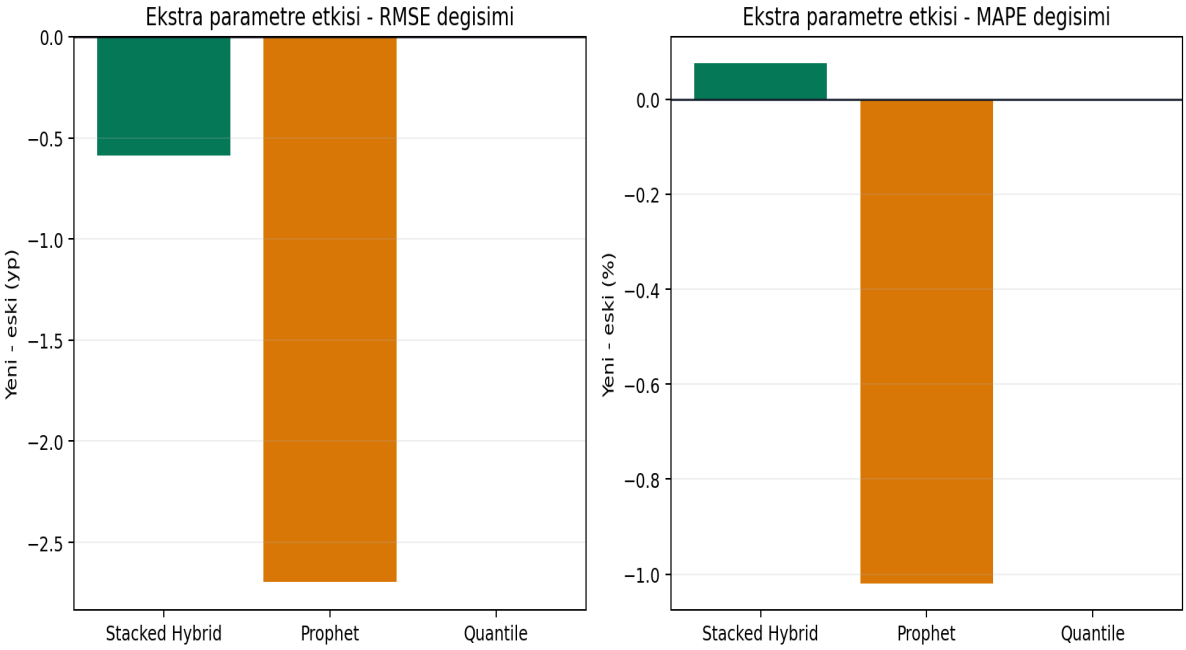
Gorsel: extra_param_models_holdout.png




Gorsel: extra_param_models_future.png




Gorsel: extra_param_improvement_vs_previous.png


Ekstra parametre etkisi - RMSE degisimi
Yeni - eski (yp)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
Stacked Hybrid
Prophet
Quantile
Ekstra parametre etkisi - MAPE degisimi
Yeni - eski (%)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
Stacked Hybrid
Prophet
Quantile